ANOVA (one-way)

Function:
It calculates: SSbet, SSwith, dfbet, dfwith, MSbet, MSwith, F, pvalue, partial eta squared, and statistical significance.

```
def anova(*groups, group_names=None, alpha=0.05):
    g = len(groups)
    N = sum(len(group) for group in groups)

    if group_names is None:
        labels = [f"Group {i + 1}" for i in range(g)]
    else:
        labels = group_names

    group_means = []
    group_ss = []

    print("\nGroup Statistics")
    print(f"{'Group':<15} {'n':>5} {'M':>10} {'SD':>10} {'SS':>10}")
    print("-" * 55)

    for i in range(g):
        group = groups[i]

        M = mean(group)
        SS = ss(group)
        SD = std_dev(group)

        group_means.append(M)
        group_ss.append(SS)

        print(f"{labels[i]:<15} {len(group):>5} {M:>10.3f} {SD:>10.3f} {S

    SSwithin = sum(group_ss)

    n = len(groups[0])

    sum_means_squared = sum(M ** 2 for M in group_means)
    sum_means = sum(group_means)

    SSbetween = n * (sum_means_squared - ((sum_means ** 2) / g))

    SStotal = SSbetween + SSwithin

    dfbetween = g - 1
    dfwithin = N - g
    dftotal = N - 1
```

```
    MSbetween = SSbetween / dfbetween
    MSwithin = SSwithin / dfwithin

    F = MSbetween / MSwithin

    pvalue = 1 - stats.f.cdf(F, dfbetween, dfwithin)

    eta_sq = SSbetween / (SSbetween + SSwithin)

    print("\nANOVA Source Table")
    print("-" * 75)
    print(f"{'Source':<15} {'SS':>10} {'df':>8} {'MS':>10} {'F':>10} {'p
    print("-" * 75)
    print(f"{'Between':<15} {SSbetween:>10.3f} {dfbetween:>8} {MSbetween:
    print(f"{'Within':<15} {SSwithin:>10.3f} {dfwithin:>8} {MSwithin:>10.
    print(f"{'Total':<15} {SStotal:>10.3f} {dftotal:>8}")
    print("-" * 75)

    print(f"\nPartial eta squared = {eta_sq:.4f}")

    # Tukey HSD

    if pvalue < alpha:
        print(f"\nThe ANOVA is significant because p = {pvalue:.4f} < {a
        print("\nTukey HSD post hoc test:")

        tukey = stats.tukey_hsd(*groups)

        print(f"{'Comparison':<25} {'Mean Diff':>12} {'p-value':>12} {'Si
        print("-" * 70)

        for i in range(g):
            for j in range(i + 1, g):
                difference = tukey.statistic[i][j]
                p = tukey.pvalue[i][j]

                if p < alpha:
                    significant = "Yes"
                else:
                    significant = "No"

                comparison = f"{labels[i]} - {labels[j]}"

                print(f"{comparison:<25} {difference:>12.3f} {p:>12.4f} {
```

```
    else:
        print(f"\nThe ANOVA is not significant because p = {pvalue:.4f} >
        print("Tukey HSD is not needed.")

    return SSbetween, SSwithin, dfbetween, dfwithin, MSbetween, MSwithin,
```

2. A Research team is trying to find out if different types of drinks given before an exam affect exam scores. A research team split students into 3 random groups and gave them cola, club soda, or water. Did the type of drink affect their exam scores?

```
# Question 2: Drinks example

cola = [88, 90, 92]
club_soda = [83, 85, 87]
water = [80, 82, 84]

anova(
    cola,
    club_soda,
    water,
    group_names=["Cola", "Club Soda", "Water"]
)
```

After running it I got:

```
Group Statistics
Group              n          M         SD         SS
-------------------------------------------------------
Cola               3     90.000      2.000      8.000
Club Soda          3     85.000      2.000      8.000
Water              3     82.000      2.000      8.000

ANOVA Source Table
-------------------------------------------------------------------------------
Source                  SS       df         MS          F          p
-------------------------------------------------------------------------------
Between             98.000        2     49.000     12.250     0.0076
Within              24.000        6      4.000
Total              122.000        8
-------------------------------------------------------------------------------

Partial eta squared = 0.8033

The ANOVA is significant because p = 0.0076 < 0.05.

Tukey HSD post hoc test:
Comparison                  Mean Diff      p-value     Significant
-----------------------------------------------------------------
Cola - Club Soda                5.000       0.0504              No
Cola - Water                    8.000       0.0065             Yes
Club Soda - Water               3.000       0.2366              No
```

The type of drink given before the exam had a significant effect on exam scores, $F(2, 6) = 12.25$, $p = .0076$, partial $\eta^2 = .8033$. Therefore, the null hypothesis is rejected. The Tukey HSD test showed that the Cola group scored significantly higher than the Water group, but Cola and Club Soda were not significantly different, and Club Soda and Water were not significantly different. Overall, students who drank cola had higher exam scores than students who drank water.

3. Challenge problem:

```
# Challenge: Page 288 example

placebo = [8, 9, 6, 9, 9]
mg100 = [7, 9, 8, 10, 8]
mg250 = [7, 6, 8, 7, 9]
mg500 = [4, 5, 6, 5, 7]

anova(
    placebo,
    mg100,
    mg250,
    mg500,
    group_names=["Placebo", "100 mg", "250 mg", "500 mg"]
)
```

So I got:

```
Group Statistics
Group               n          M         SD         SS
-------------------------------------------------------
Placebo             5      8.200      1.304      6.800
100 mg              5      8.400      1.140      5.200
250 mg              5      7.400      1.140      5.200
500 mg              5      5.400      1.140      5.200

ANOVA Source Table
-------------------------------------------------------------------------------
Source                   SS       df         MS          F          p
-------------------------------------------------------------------------------
Between              28.150        3      9.383      6.702     0.0039
Within               22.400       16      1.400
Total                50.550       19
-------------------------------------------------------------------------------

Partial eta squared = 0.5569

The ANOVA is significant because p = 0.0039 < 0.05.

Tukey HSD post hoc test:
Comparison                   Mean Diff      p-value     Significant
-----------------------------------------------------------------------
Placebo - 100 mg                -0.200       0.9931              No
Placebo - 250 mg                 0.800       0.7125              No
Placebo - 500 mg                 2.800       0.0087             Yes
100 mg - 250 mg                  1.000       0.5545              No
100 mg - 500 mg                  3.000       0.0050             Yes
250 mg - 500 mg                  2.000       0.0714              No
```

The ANOVA was significant, $F(3, 16) = 6.70$, $p = .0039$, partial $\eta^2 = .5569$. Therefore, the null hypothesis is rejected, and at least one treatment group mean was significantly different from another.The Tukey HSD test showed that the 500 mg group was significantly different from both the Placebo group and the 100 mg group. However, the other group comparisons were not significant.